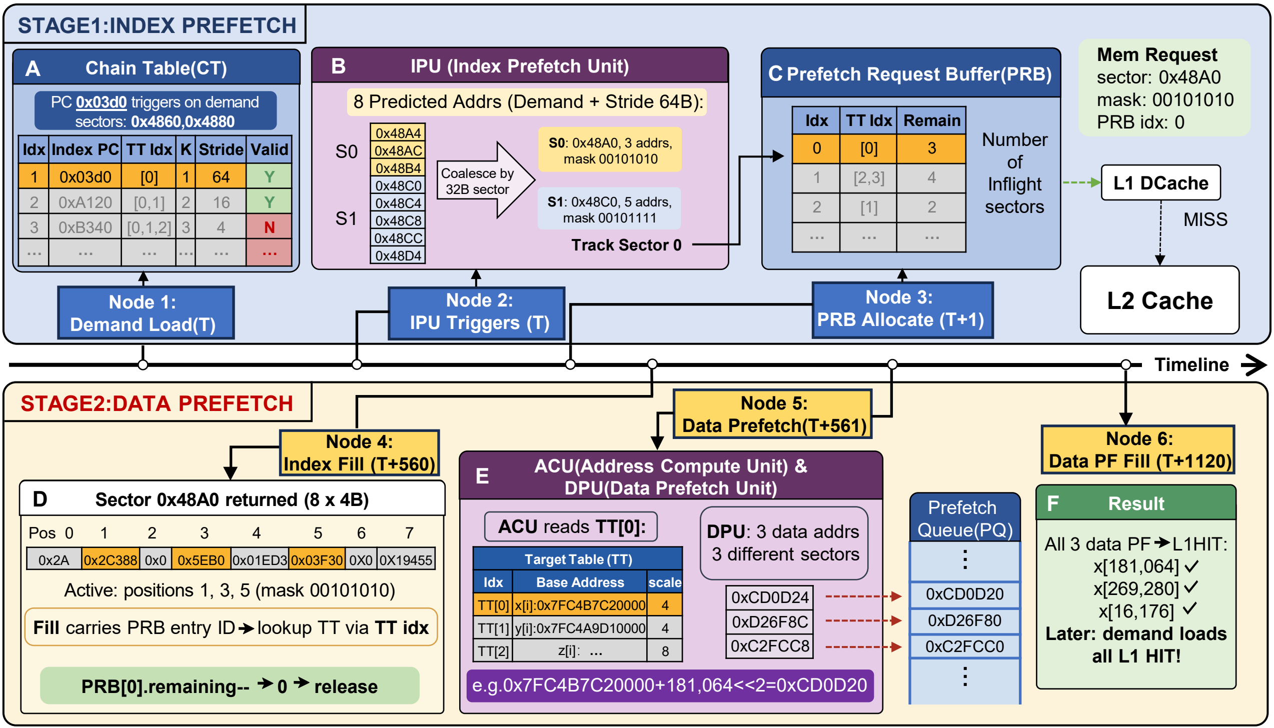

STAGE1:INDEX PREFETCH
A Chain Table(CT)
PC 0x03d0 triggers on demand sectors: 0x4860,0x4880
Idx | Index PC | TT Idx | K | Stride | Valid
1 | 0x03d0 | [0] | 1 | 64 | Y
2 | 0xA120 | [0,1] | 2 | 16 | Y
3 | 0xB340 | [0,1,2] | 3 | 4 | N
B IPU (Index Prefetch Unit)
8 Predicted Addrs (Demand + Stride 64B):
S0
S1
0x48A4
0x48AC
0x48B4
0x48C0
0x48C4
0x48C8
0x48CC
0x48D4
Coalesce by 32B sector
S0: 0x48A0, 3 addrs, mask 00101010
S1: 0x48C0, 5 addrs, mask 00101111
Track Sector 0
C Prefetch Request Buffer(PRB)
Idx | TT Idx | Remain
0 | [0] | 3
1 | [2,3] | 4
2 | [1] | 2
Number of Inflight sectors
Mem Request
sector: 0x48A0
mask: 00101010
PRB idx: 0
L1 DCache
MISS
L2 Cache
Node 1: Demand Load(T)
Node 2: IPU Triggers (T)
Node 3: PRB Allocate (T+1)
Timeline
STAGE2:DATA PREFETCH
Node 4: Index Fill (T+560)
Node 5: Data Prefetch(T+561)
Node 6: Data PF Fill (T+1120)
D Sector 0x48A0 returned (8 x 4B)
Pos 0 1 2 3 4 5 6 7
0x2A 0x2C388 0x0 0x5EB0 0x01ED3 0x03F30 0X0 0X19455
Active: positions 1, 3, 5 (mask 00101010)
Fill carries PRB entry ID → lookup TT via TT idx
PRB[0].remaining-- → 0 → release
E ACU(Address Compute Unit) & DPU(Data Prefetch Unit)
ACU reads TT[0]:
Target Table (TT)
Idx | Base Address | scale
TT[0] | x[i]:0x7FC4B7C20000 | 4
TT[1] | y[i]:0x7FC4A9D10000 | 4
TT[2] | z[i]: ... | 8
DPU: 3 data addrs 3 different sectors
0xCD0D24
0xD26F8C
0xC2FCC8
e.g.0x7FC4B7C20000+181,064<<2=0xCD0D20
Prefetch Queue(PQ)
0xCD0D20
0xD26F80
0xC2FCC0
F Result
All 3 data PF → L1HIT:
x[181,064] ✓
x[269,280] ✓
x[16,176] ✓
Later: demand loads all L1 HIT!